

Name _____

Date _____

Springs, Solids and Fluids: Forces You Have to Measure

SYNOPSIS

Nine notes of this module have done the same thing: pick an object, draw the forces on it, apply $\Sigma F = ma$. This one is different in kind. It introduces three forces — a stretched spring, a squeezed solid, the air pushing back on a falling body — whose strengths **do not follow from Newton's laws at all**. Each carries a constant that nobody can derive and everybody has to measure. That is not a gap in the theory. It is how physics meets real materials.

BEFORE YOU READ ON — PREDICT

A skydiver steps out of an aircraft. From **N4** you know that near the Earth every falling object accelerates at 9.8 m/s^2 , whatever its mass.

Does the skydiver still accelerate at 9.8 m/s^2 ten seconds later? Thirty?

Answer, with one sentence saying what would have to be true for your answer to be right, before you read on.

PERFORMANCE EXPECTATIONS

1. **Relate** the force in a stretched or compressed spring to its extension, and **distinguish** the extension from the spring's total length.
2. **Determine** k from the slope of a force-against-extension graph, state its unit, and **interpret** it as stiffness.
3. **Apply** the stress-strain relations for compression, stretch and shear, and **say** what Y and S describe.
4. **Explain** how terminal velocity arises from a force that grows with speed competing with one that does not.
5. **Compute** a terminal velocity from mass, area, shape and fluid density.

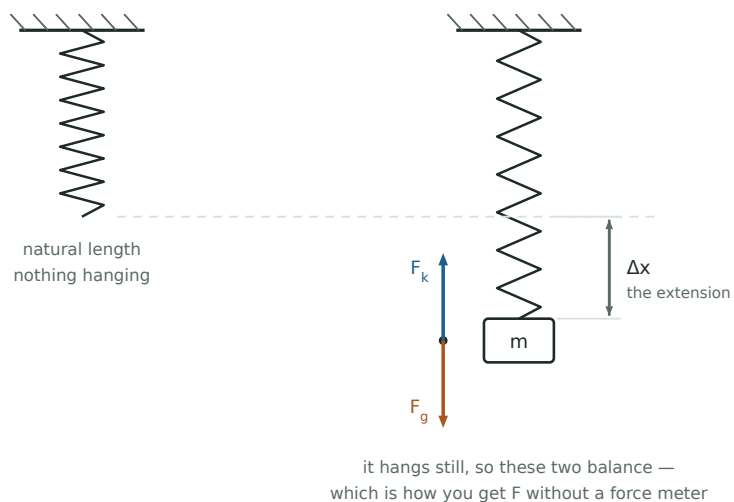
6. **Explain** why k , Y , S and C_D have to be measured rather than derived, and **name** the constant from earlier in this module that is the same kind of quantity.

MOVEMENT 1 — A DIFFERENT KIND OF FORCE LAW

Laws you derive, and laws you measure

MOVEMENT 2 — SPRINGS

Hooke's law



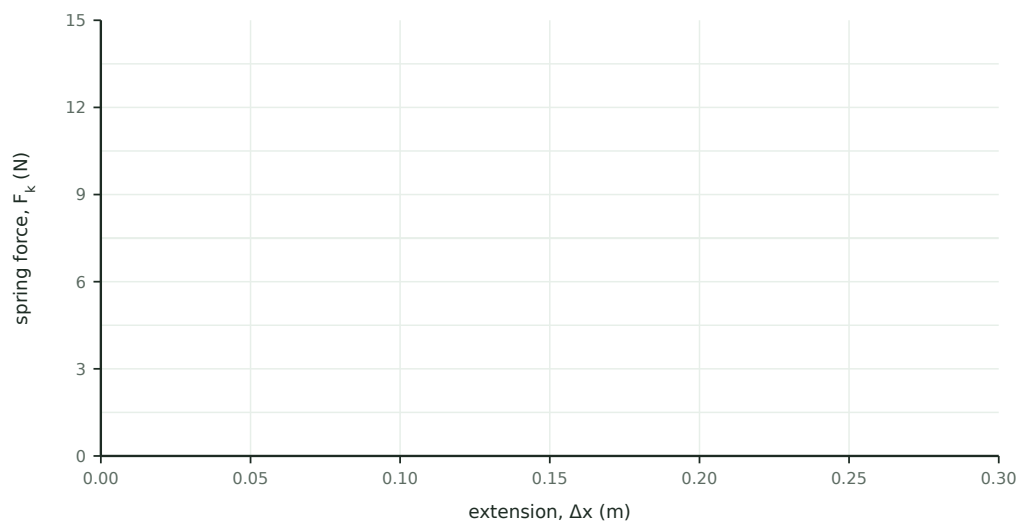
Eq. 1 $F_k = k\Delta x$

double the extension and the spring pulls back twice as hard

Measuring k

Hang known weights, record the extension each time, and plot one against the other. If Hooke's law is right the points fall on a straight line through the origin, and the slope of that line is the spring constant.

HANGING WEIGHT, F (N)	EXTENSION, Δx (CM)	Δx IN METRES
2		
4		
12		



Plot force against extension and draw the best straight line. Note the units on the horizontal axis — convert before you plot, not after.

1. Find the slope from your line, and give its unit.

2. You have two springs. One has $k = 50 \text{ N/m}$ and the other $k = 500 \text{ N/m}$. Which is harder to stretch, and by what factor? Sketch both on the same axes.

3. A spring with a larger k gives a steeper line. Say in one sentence what “steeper” means physically here.

WITH CALCULUS

You found k as the slope of a line you drew by eye. Written exactly, that slope is a derivative:

$$k = dF_k / d(\Delta x)$$

Which is a stronger statement than it looks, because a derivative is defined at a *point* and your line was drawn across a whole range.

1. If the F – Δx graph really is a straight line, what is true about $dF_k/d(\Delta x)$ at every point on it? That statement is what “ k is a constant” actually means.

2. Sketch the graph for a spring pulled well past its elastic limit. Mark the stretch where $dF_k/d(\Delta x)$ is constant, and say what is happening to that derivative beyond it.

3. So state the elastic limit as a condition on the **derivative** rather than on the spring.

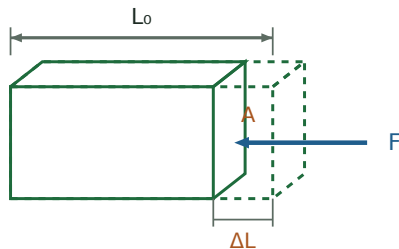
4. Y and S were defined the same way, as stress over strain. Write each as a derivative. What plays the part of Δx in each one?

5. A constant obtained this way — the slope of a curve across the stretch where the curve is near enough straight — is called a **local linearization**. Using your sketch from task 2, say what the word *local* is doing in that phrase.

MOVEMENT 3 — SOLIDS

Hooke's law, spread over an area

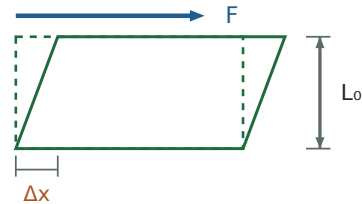
COMPRESSION AND STRETCH



A is the face the force pushes on

$$F / A = Y (\Delta L / L_0)$$

SHEAR



the top face slides; the block leans

$$F / A = S (\Delta x / L_0)$$

IN BOTH CASES

stress = modulus $\times$ strain

Y is the **Young's modulus** and S the **shear modulus**. Both are properties of the material, not of the particular block — steel has the same Y whatever shape you cut it into, which is exactly why writing the law in terms of F/A and $\Delta L/L_0$ rather than F and ΔL was worth the trouble.

1. Compare $F = k\Delta x$ with $F/A = Y(\Delta L/L_0)$. Which symbol in the second plays the part that k plays in the first? What did the other two changes buy you?
2. Two bars of the same steel, same length, one twice the cross-sectional area. Under the same force, which shortens more, and by what factor?

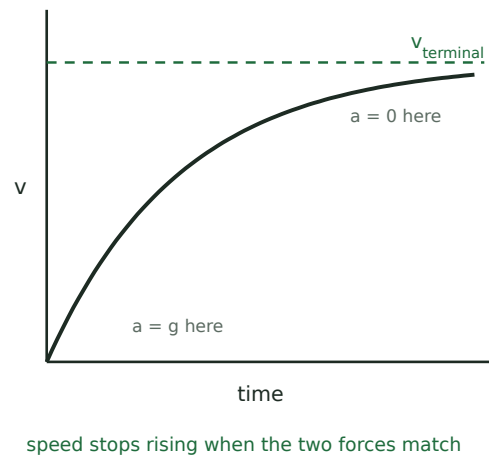
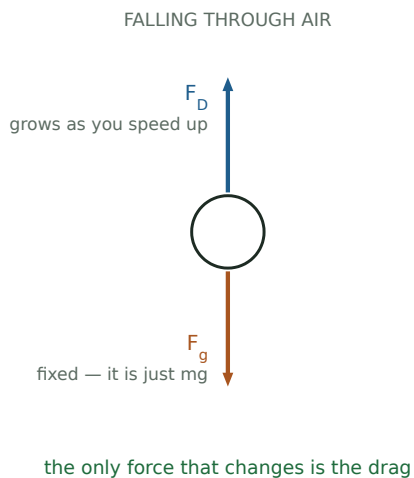
3. A material with a large γ — is it stiff or floppy? Give an example of each.

MOVEMENT 4 — FLUIDS

Drag, and why falling stops speeding up

Eq. 1 $F_D = \frac{1}{2} C_D \rho A v^2$

C_D shape $\cdot \rho$ fluid density $\cdot A$
frontal area



4. Write $\Sigma F = ma$ for the falling body, taking down as positive, with both forces in it.

5. Set the acceleration to zero and solve for v . This is the terminal velocity.

Eq. 1 $v_{\text{term}} = \sqrt{\frac{2mg}{C_D \rho A}}$

the speed at which drag has
grown to equal the weight

TASK 6 — A SKYDIVER

1. Take $m = 100 \text{ kg}$, frontal area $A = 2 \text{ m}^2$, air density $\rho = 1.3 \text{ kg/m}^3$, and $C_D = 0.7$ for a spread-eagled body. Find the terminal velocity in m/s.

2. Convert it to km/h.

3. Skydivers report roughly 200 km/h. Yours will not match. Is the disagreement small enough to call this model useful? Name **one** number in the calculation you would want to know better.

4. The same skydiver goes head-down, cutting A roughly in half. What happens to the terminal velocity — and by what factor, exactly?

WITH CALCULUS

Everything so far has been about the *end* of the fall. The hook asked about the middle of it — whether the skydiver is still gaining speed after ten seconds, and after thirty. Answering that means writing the second law as an equation in dv/dt and solving it.

1. Take **down as positive**. Write Newton's second law for the falling skydiver with drag included, as an equation containing dv/dt .

2. Set $dv/dt = 0$ and solve for v . Compare with the v_{term} you used in Task 6. What have you just shown about where that formula came from?

3. **Separate the variables and integrate**, taking $v = 0$ at $t = 0$, to show that $v(t) = v_{\text{term}} \tanh(gt / v_{\text{term}})$. Write your v -integral in terms of v_{term} first — the standard form is easier to recognise that way.

4. Check your solution at the two ends: what is v at $t = 0$, and what does v approach as t grows?

5. v_{term}/g has units of time. Work it out from your own Task 6 answer. Roughly how many of those does the fall need to reach 99% of terminal velocity?

6. Sketch v against t , marking v_{term} and your timescale on the axes. Now answer the question in the hook from the graph rather than in words: still accelerating at ten seconds? At thirty?

BACK TO YOUR PREDICTION

You predicted whether the skydiver is still accelerating at 9.8 m/s^2 after ten seconds and after thirty. Was it right? Point to the moment on the speed-against-time graph where your answer changes, and name the force that changed it.
